Lab 0\L0q2.java

```java
//2.    Write a program in Java to generate first n prime numbers.

import java.io.*;
import java.util.*;

public class L0q2 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("========================================");
        System.out.println("Name: Patel Ishan");
        System.out.println("Roll Number: 25BCP113");
        System.out.println("========================================");

        System.out.println("Program: Generate the first 'n' prime numbers.");
        System.out.println("----------------------------------------");

        System.out.print("Please enter the total number of prime numbers you want to generate: ");
        int n = sc.nextInt();

        System.out.println("\nGenerating the first " + n + " prime numbers:");
        System.out.println("----------------------------------------");

        int m = 2;
        for(int i = 1; i <= n; i++){
            while(true){
                int count = 0;
                for(int j = 1; j <= m; j++){
                    if(m % j == 0)
                        count++;
                }
                if (count > 2){
                    m++;
                    continue;
                } else {
                    // Refined output sentence
                    System.out.println("Prime Number " + i + ": " + m);
                    m++;
                    break;
                }
            }
        }

        System.out.println("----------------------------------------");
        sc.close();
    }
}

```

```
C:\JAVA_Lab\Lab 0\bin>java L0q2
========================================
Name: Patel Ishan
Roll Number: 25BCP113
========================================
Program: Generate the first 'n' prime numbers.
----------------------------------------
Please enter the total number of prime numbers you want to generate: 5

Generating the first 5 prime numbers:
----------------------------------------
Prime Number 1: 2
Prime Number 2: 3
Prime Number 3: 5
Prime Number 4: 7
Prime Number 5: 11
----------------------------------------
```

Lab 0\L0q3.java

```java
//3.    Write a program to enter two numbers and perform mathematical operations on them.

import java.io.*;
import java.util.*;

public class L0q3 {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System.out.println("========================================");
      System.out.println("Roll Number: 25BCP113");
      System.out.println("Name: Patel Ishan");
      System.out.println("========================================");

      System.out.println("Program: Perform basic mathematical operations on two numbers.");
      System.out.println("----------------------------------------");

      System.out.print("Please enter the first number: ");
      double a = sc.nextDouble();

      System.out.print("Please enter the second number: ");
      double b = sc.nextDouble();

      System.out.println("\nCalculation Results:");
      System.out.println("----------------------------------------");

      System.out.println("The Sum of the numbers is: " + (a + b));
      System.out.println("The Difference (First - Second) is: " + (a - b));
      System.out.println("The Product (Multiplication) is: " + (a * b));

      if (b != 0) {
         System.out.println("The Quotient (First / Second) is: " + (a / b));
      } else {
         System.out.println("The Quotient (First / Second) is: Undefined (Cannot divide by zero)");
      }

      System.out.println("----------------------------------------");
      sc.close();
    }
}

```

```
C:\JAVA_Lab\Lab 0\bin>java L0q3
==========================================
Roll Number: 25BCP113
Name: Patel Ishan
==========================================
Program: Perform basic mathematical operations on two numbers.
------------------------------------------
Please enter the first number: 15
Please enter the second number: 10

Calculation Results:
------------------------------------------
The Sum of the numbers is: 25.0
The Difference (First - Second) is: 5.0
The Product (Multiplication) is: 150.0
The Quotient (First / Second) is: 1.5
------------------------------------------
```

Lab 0\L0q4.java

```java
// 4. Write a program that calculates percentage marks of the student if marks of 6 subjects are given.

import java.io.*;
import java.util.*;

public class L0q4 {
    public static class Student {
        public int rNo;
        public String name;
        public int marks[] = new int[6];

        public void getData() {
            Scanner sc = new Scanner(System.in);

            System.out.print("Enter the student's Roll Number: ");
            rNo = sc.nextInt();

            sc.nextLine();

            System.out.print("Enter the student's Name: ");
            name = sc.nextLine();

            System.out.println("\nPlease enter the marks for 6 subjects:");
            for(int i = 0; i < marks.length; i++) {
                System.out.print("Enter the marks for Subject " + (i + 1) + ": ");
                marks[i] = sc.nextInt();
            }
        }

        public void disp() {
            System.out.println("\n--- Student Details ---");
            System.out.println("Roll Number: " + rNo);
            System.out.println("Student Name: " + name);

            int sum = 0;
            for(int i = 0; i < marks.length; i++) {
                sum = sum + marks[i];
            }

            float percentage = (float) sum / marks.length;

            System.out.println("Total Marks: " + sum);
            System.out.println("Percentage: " + percentage + "%");
        }
    }

    public static void main(String[] args) {

        System.out.println("========================================");
        System.out.println("Roll Number: 25BCP113");
        System.out.println("Name: Patel Ishan");
        System.out.println("========================================");

        System.out.println("This program calculates a student's total marks and percentage based on 6 subjects.\n");
```

```
55          System.out.println("----------------------------------------");
56          Student s1 = new Student();
57          s1.getData();
58          s1.disp();
59          System.out.println("----------------------------------------");
60      }
61  }
```

```
C:\JAVA_Lab\Lab 0\bin>java L0q4
========================================
Roll Number: 25BCP113
Name: Patel Ishan
========================================
This program calculates a student's total marks and percentage based on 6 subjects.

----------------------------------------
Enter the student's Roll Number: 113
Enter the student's Name: Ishan

Please enter the marks for 6 subjects:
Enter the marks for Subject 1: 56
Enter the marks for Subject 2: 68
Enter the marks for Subject 3: 78
Enter the marks for Subject 4: 96
Enter the marks for Subject 5: 63
Enter the marks for Subject 6: 86

--- Student Details ---
Roll Number: 113
Student Name: Ishan
Total Marks: 447
Percentage: 74.5%
----------------------------------------
```

Lab 0\L0q5.java

```java
// 5. Write a program in Java to find maximum of three numbers using conditional operator.

import java.io.*;
import java.util.*;

public class L0q5 {
    public static void main(String[] args) {
        System.out.println("========================================");
        System.out.println("Name: Patel Ishan");
        System.out.println("Roll Number: 25BCP113");
        System.out.println("========================================");

        System.out.println("This program finds the maximum of three numbers using the conditional operator.\n");
        System.out.println("----------------------------------------");

        Scanner sc = new Scanner(System.in);

        System.out.println("Let's find the maximum of three numbers.");

        System.out.print("Please enter the first number: ");
        int a = sc.nextInt();

        System.out.print("Please enter the second number: ");
        int b = sc.nextInt();

        System.out.print("Please enter the third number: ");
        int c = sc.nextInt();

        int max = (a > b) ? ((a > c) ? a : c) : ((b > c) ? b : c);

        System.out.println("\nThe maximum number among " + a + ", " + b + ", and " + c + " is: " + max);
        System.out.println("----------------------------------------");

        sc.close();
    }
}
```

```
C:\JAVA_Lab\Lab 0\bin>java L0q5
=======================================
Name: Patel Ishan
Roll Number: 25BCP113
=======================================
This program finds the maximum of three numbers using the conditional operator.

---------------------------------------
Let's find the maximum of three numbers.
Please enter the first number: 15
Please enter the second number: 20
Please enter the third number: 13

The maximum number among 15, 20, and 13 is: 20
---------------------------------------
```

Lab 0\L0q6.java

```java
// 6. Write a program to accept a line and check how many consonants and vowels are there in line.

import java.io.*;
import java.util.*;

public class L0q6 {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.println("========================================");
        System.out.println("Name: Patel Ishan");
        System.out.println("Roll Number: 25BCP113");
        System.out.println("========================================");

        System.out.println("This program counts the number of vowels and consonants in a given line of text.\n");
        System.out.println("----------------------------------------");

        System.out.print("Please enter a sentence or line of text: ");
        String line = scanner.nextLine();

        int vowelsCount = 0;
        int consonantsCount = 0;

        String lowerCaseLine = line.toLowerCase();

        for (int i = 0; i < lowerCaseLine.length(); i++) {
            char ch = lowerCaseLine.charAt(i);

            if (Character.isLetter(ch)) {
                if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
                    vowelsCount++;
                } else {
                    consonantsCount++;
                }
            }
        }

        System.out.println("\n--- Results ---");
        System.out.println("Total Vowels found: " + vowelsCount);
        System.out.println("Total Consonants found: " + consonantsCount);
        System.out.println("----------------------------------------");

        scanner.close();
    }
}
```

```
C:\JAVA_Lab\Lab 0\bin>java L0q6
========================================
Name: Patel Ishan
Roll Number: 25BCP113
========================================
This program counts the number of vowels and consonants in a given line of text.

----------------------------------------
Please enter a sentence or line of text: Hello World Ishan Here

--- Results ---
Total Vowels found: 7
Total Consonants found: 12
----------------------------------------
```

Lab 0\L0q7.java

```java
// 7. Write a program to count the number of words that start with capital letters.

import java.io.*;
import java.util.*;

public class L0q7 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("========================================");
        System.out.println("Name: Patel Ishan");
        System.out.println("Roll Number: 25BCP113");
        System.out.println("========================================");

        System.out.println("This program counts the number of words that start with a capital letter.\n");
        System.out.println("----------------------------------------");

        System.out.print("Please enter a sentence: ");
        String s1 = sc.nextLine();

        String s2[] = s1.split("\\s+");

        int count = 0;

        for(int i = 0; i < s2.length; i++){
            String temp = s2[i];

            if(temp.length() > 0) {
                char x = temp.charAt(0);

                if(x >= 65 && x <= 90) {
                    count++;
                }
            }
        }

        System.out.println("\n--- Results ---");
        System.out.println("Number of words that start with a capital letter: " + count);
        System.out.println("----------------------------------------");

        sc.close();
    }
}
```

```
C:\JAVA_Lab\Lab 0\bin>java L0q7
========================================
Name: Patel Ishan
Roll Number: 25BCP113
========================================
This program counts the number of words that start with a capital letter.

----------------------------------------
Please enter a sentence: Hello ishan Here How are You

--- Results ---
Number of words that start with a capital letter: 4
----------------------------------------
```

Lab 0\L0q8.java

```java
// 8. Create a class which ask the user to enter a sentence, and it should display count of each vowel type in the sentence.
//    The program should continue till user enters a word “quit”. Display the total count of each vowel for all sentences.

import java.io.*;
import java.util.*;

public class L0q8 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("========================================");
        System.out.println("Name: Patel Ishan");
        System.out.println("Roll Number: 25BCP113");
        System.out.println("========================================");
        System.out.println("This program counts how many of each vowel are used. Type 'quit' at the end to stop.\n");
        System.out.println("----------------------------------------");

        int countA = 0, countE = 0, countI = 0, countO = 0, countU = 0;

        System.out.print("Please enter a sentence (end with 'quit'): ");

        while (true) {
            String s1 = sc.next();
            String s2 = s1.toLowerCase();

            if (s2.equals("quit")) {
                break;
            }

            for (int i = 0; i < s2.length(); i++) {
                char x = s2.charAt(i);

                switch (x) {
                    case 'a': countA++; break;
                    case 'e': countE++; break;
                    case 'i': countI++; break;
                    case 'o': countO++; break;
                    case 'u': countU++; break;
                }
            }
        }

        System.out.println("\n--- Final Vowel Counts ---");
        System.out.println("A: " + countA);
        System.out.println("E: " + countE);
        System.out.println("I: " + countI);
        System.out.println("O: " + countO);
        System.out.println("U: " + countU);

        System.out.println("----------------------------------------");
        sc.close();
    }
}
```

```
C:\JAVA_Lab\Lab 0\bin>java L0q8
========================================
Name: Patel Ishan
Roll Number: 25BCP113
========================================
This program counts how many of each vowel are used. Type 'quit' at the end to stop.

----------------------------------------
Please enter a sentence (end with 'quit'): Hyy my name is ishan patel quit

--- Final Vowel Counts ---
A: 3
E: 2
I: 2
O: 0
U: 0
----------------------------------------
```

Lab 0\L0q9.java

```java
//9.    Write an interactive program to print a string entered in a pyramid form. For instance, the string “stream” has to be displayed as follows:
//               S
//              S t
//             S t r
//            S t r e
//           S t r e a
//          S t r e a m

import java.io.*;
import java.util.*;

public class L0q9 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("========================================");
        System.out.println("Name: Patel Ishan");
        System.out.println("Roll Number: 25BCP113");
        System.out.println("========================================");

        System.out.println("This program takes a word from the user and prints it in a pyramid pattern.\n");
        System.out.println("----------------------------------------");

        System.out.print("Please enter a word: ");
        String s = sc.next();
        int n = s.length();

        System.out.println();

        for(int i = 1; i <= n; i++){
            for(int j = 1; j <= (n - i); j++){
                System.out.print(" ");
            }
            for(int k = 0; k < i; k++){
                System.out.print(s.charAt(k) + " ");
            }
            System.out.println();
        }
        System.out.println("----------------------------------------");
        sc.close();
    }
}

```

```
C:\JAVA_Lab\Lab 0\bin>java L0q9
========================================
Name: Patel Ishan
Roll Number: 25BCP113
========================================
This program takes a word from the user and prints it in a pyramid pattern.

----------------------------------------
Please enter a word: Ishan

    I
   I s
  I s h
 I s h a
I s h a n
----------------------------------------
```

Lab 0\L0q10.java

```java
//10.   Write an interactive program to print a diamond shape. For example, if user enters the number 3, the diamond will be as follows:
//      *
//     * *
//    * * *
//     * *
//      *

import java.io.*;
import java.util.*;

public class L0q10 {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("========================================");
        System.out.println("Name: Patel Ishan");
        System.out.println("Roll Number: 25BCP113");
        System.out.println("========================================");

        System.out.println("This program prints a diamond shape of stars based on the number you enter.\n");
        System.out.println("----------------------------------------");

        System.out.print("Please enter the maximum number of stars for the middle row: ");
        int n = sc.nextInt();

        System.out.println();

        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= n - i; j++) {
                System.out.print(" ");
            }
            for (int j = 1; j <= i; j++) {
                System.out.print("* ");
            }
            System.out.println();
        }

        for (int i = n - 1; i >= 1; i--) {
            for (int j = 1; j <= n - i; j++) {
                System.out.print(" ");
            }
            for (int j = 1; j <= i; j++) {
                System.out.print("* ");
            }
            System.out.println();
        }
        System.out.println("----------------------------------------");
        sc.close();
    }
}
```

```
C:\JAVA_Lab\Lab 0\bin>java L0q10
========================================
Name: Patel Ishan
Roll Number: 25BCP113
========================================
This program prints a diamond shape of stars based on the number you enter.

----------------------------------------
Please enter the maximum number of stars for the middle row: 5

    *
   * *
  * * *
 * * * *
* * * * *
 * * * *
  * * *
   * *
    *
----------------------------------------
```